Jehyeok Yeon

Incoming Ph.D. Candidate at Max Planck Institute for Intelligent Systems

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Research Interests

My research interest is to develop frameworks for **scalable oversight** and evaluation of increasingly capable AI systems, particularly in areas where models can exceed human competence. I aim to formalize and measure the **limits of human supervision**, and to design **certification-style guarantees and analysis methods** that remain meaningful even as model architectures and capabilities evolve. My current research focuses include:

- **Epistemic Limits and Alignment under Uncertainty:** Studying how pluralistic human values, incomplete knowledge, and scientific uncertainty constrain alignment objectives, and developing frameworks that handle conflict, uncertainty, and shifting world models.
- **Certification-Oriented Safety Metrics:** Moving beyond single-score static benchmarks toward statistical guarantees, worst-case analysis, and certificates that remain stable under scaling and distribution shift.

Education

Ph.D. in Computer Science July 2026 – Current
Max Planck Institute for Intelligent Systems
Advisor: Maksym Andriushchenko

BS in Computer Science + Linguistics August 2022 – December 2025
University of Illinois Urbana-Champaign (UIUC) GPA 3.92/4.0
Advisor: Prof. Gagandeep Singh

Selected Publications

- **Jehyeok Yeon,** Hyeonjeong Ha, Qiusi Zhan, Heng Ji. *Securing Multimodal AI through Internal Information Decomposition.* **ICML 2026 (Spotlight).**
- **Jehyeok Yeon**, Ben Rank, Maksym Andriushchenko. *InferenceBench: A Benchmark for Open-Ended LLM Inference Optimization by AI Agents*. **ICML 2026 AIWILD (Spotlight)**
- **Jehyeok Yeon,** Isha Chaudhary, Gagandeep Singh. *Certifying Robustness of Agent Tool-Selection Under Adversarial Attacks.* **ICLR 2026 AIWILD**
- Hangoo Kang*, **Jehyeok Yeon***, Gagandeep Singh. *TRAP: Targeted Redirecting of Agentic Preferences.* **NeurIPS 2025.** (*indicates equal contribution)

- Ben Rank*, Lena Libon*, **Jehyeok Yeon**, Jeremy Qin, David Schmotz, Derck Prinzhorn, Daniel Donnelly, Maksym Andriushchenko. *ResearchArena: Evaluating Sabotage and Monitoring in Automated AI R&D.* (Under Review) (*indicates equal contribution)

Research Experience

London AI Safety Research Labs July 2026 – October 2026

Research Scholar with Cozmin Ududec

- Incoming cohort to work on Science of Evaluations with UK AI Security Institute.

Max Planck Institute for Intelligent Systems January 2026 – June 2026

Visiting Student Researcher with Maksym Andriushchenko

- Conducted Agentic R&D research to create long-horizon benchmarks for AI optimization tasks.

University of Southern California May 2025 – June 2026

Visiting Student Researcher with Prof. Luca Luceri

- Conducted Information Geometry research to model and analyze LLM internals as complex, non-linear data structures.

University of Illinois Urbana-Champaign August 2024 – June 2026

Research Assistant with Prof. Gagandeep Singh and Prof. Heng Ji

- Conducted quantitative certification research to model and evaluate the safety of agentic AI systems.
- Conducted AI Security research to develop a novel indirect prompt injection method utilizing semantic injection on images.

Work Experience

Google DeepMind June 2026 – Current

External Research Advisor

- Collaborating with Core Data and Eval team on evaluation methodology, benchmark design, and research direction related to long-horizon agentic capability under NDA.

Intradiem May 2025 – August 2025

Machine Learning Intern

- Reduced false-negative burnout classifications by 35% by building feature extraction pipelines with PyTorch, Temporal, and distributed SQL.
- Built predictive models to detect burnout anomalies across terabytes of call center telemetry data.

Hanwha Life Insurance June 2024 – August 2024

AI Data Scientist

- Deployed a RAG Agent utilizing LangChain, ChromaDB, and fine-tuned instruction models.

- Optimized hybrid retrieval and prompt orchestration to improve average query accuracy by 40% while maintaining resolution overhead in domain-specific QA tasks.

ATLAS May 2023 – August 2024

Machine Learning Intern

- Engineered a real-time predictive model utilizing hyper-local streaming sensor data for the city of Chicago, enhancing the city's resource allocation and optimizing response times by 15% during peak events.
- Minimized inference latency for meteorological forecasting using TensorFlow Lite and ElasticSearch.

Honors and Awards

- NSF Graduate Research Fellowship Program (GRFP) (2026)
- AI4GOOD Travel Grant (2026)
- Sandia Data Challenge Finalist Award (2025)
- UIUC Get Experience Scholarship (2024)

Talks

What Open-Ended Benchmarks Reveal About AI Research Automation

- Google DeepMind; Turing; CAMEL AI; KAIST University (2026)

Rise of Indirect Prompt Injection Methods on Open Environment Agents

- AI Safety Tübingen Reading Group (2026)

Academic Service

Reviewer

- NeurIPS 2026, AI4GOOD Workshop @ ICML 2026, Agents in the Wild @ ICML 2026

Teaching Assistant

- AI Safety Course Summer 2026 @ University of Tübingen

Skills

- Languages: Python, Java, C++, SQL
- Frameworks: PyTorch, TensorFlow, Transformers, vLLM, TransformerLens, LangChain, ChromaDB, W&B, Docker, Git, Temporal, MLflow, ElasticSearch, Spring Boot

Publications

- **Jehyeok Yeon,** Hyeonjeong Ha, Qiusi Zhan, Heng Ji. *Securing Multimodal AI through Internal Information Decomposition.* **ICML 2026 (Spotlight).**

- **Jehyeok Yeon**, Ben Rank, Maksym Andriushchenko. *InferenceBench: A Benchmark for Open-Ended LLM Inference Optimization by AI Agents*. **ICML 2026 AIWILD (Spotlight)**
- **Jehyeok Yeon,** Isha Chaudhary, Gagandeep Singh. *Certifying Robustness of Agent Tool-Selection Under Adversarial Attacks*. **ICLR 2026 AIWILD**
- **Jehyeok Yeon**, Federico Cinus, Yifan Wu, Luca Luceri. *GSAE: Graph-Regularized Sparse Autoencoders for Robust LLM Safety Steering*. **IC 2026 AI4GOOD**
- Hangoo Kang*, **Jehyeok Yeon***, Gagandeep Singh. *TRAP: Targeted Redirecting of Agentic Preferences*. **NeurIPS 2025.** (*indicates equal contribution)
- **Jehyeok Yeon**, Lawrence Angrave. *The Power of Friendship: Analyzing Leadership and Adversarial Attacks in Multi-Agent Collaboration*. **ACM Collective Intelligence 2025 Poster Acceptance.**
- Ben Rank*, Lena Libon*, **Jehyeok Yeon**, Jeremy Qin, David Schmotz, Derck Prinzhorn, Daniel Donnelly, Maksym Andriushchenko. *ResearchArena: Evaluating Sabotage and Monitoring in Automated AI R&D*. (Under Review) (*indicates equal contribution)